

The dichotomy of concentration and pricing power in banking

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Abstract

In our model, banks face two informational frictions: limited pledgeability and effort moral hazard. Bankers are ex ante identical, can costlessly offer contracts to depositors, and compete in prices (deposit rates). Yet, strikingly, those bankers who attract depositors charge a mark-up which exceeds the sum of the rents that would arise from each informational friction in isolation; a positive mass of inactive bankers cannot disrupt the equilibrium despite free entry and exit. When pledgeability increases, bankers offer higher deposit rates and accept more deposits, which, in turn, leads to fewer banks (higher concentration). Thus, counterintuitively, but consistent with empirical evidence, equilibrium pricing power may decline with concentration.

Keywords: Bank concentration, Competition, Moral hazard, Pledgeability, Capacity constraints,

Pass-through

Jel codes: D40, G21, G28

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1 Introduction

Deposits are the most important funding source for banks, with the US deposit market exceeding \$18 trillion in 2025, accounting for over 80% of bank liabilities. Banks primarily compete in prices (by offering higher deposit rates to attract depositors), and yet, despite a fiercely competitive landscape comprising numerous banks (and other financial institutions), evidence shows that banks possess significant pricing power in deposit markets (see, e.g., Drechsler et al. 2017 and Corbae and D’Erasmus 2021). At the same time, empirical evidence suggests that greater bank concentration can paradoxically lead to *higher* deposit rates (see, e.g., Jackson 1992) and lower profitability (see, e.g., Berger 1995 and Claessens and Laeven 2004). Common explanations for banks’ observed pricing power include product differentiation in deposit markets and depositor switching costs. However, neither explanation alone can account for the finding that greater market concentration may actually reduce pricing power. We propose a framework in which information frictions prevent the Bertrand outcome from arising, and in which concentration and pricing power are endogenously determined in equilibrium, to provide a joint explanation for these related puzzles.

We present a model of bank competition in the deposit market which features two informational frictions: an effort moral hazard problem where bankers must exert costly and unobservable effort (after collecting deposits), and a diversion constraint which limits banks’ ability to pledge future output to secure funding in the deposit market. Despite price competition among banks in their attempt to attract deposits, the interaction of the informational constraints prevents the standard Bertrand outcome from arising – the mark-up exceeds the rent associated with either informational constraint alone. This is a particularly surprising result since we allow for (costless) free entry and exit, setting us apart from existing static models in the industrial organization literature (such as Kreps and Scheinkman 1983 and Maggi 1996, see Section 6.2 for details). Thus, our most striking theoretical result is that, even though bankers are homogeneous to begin with, and there

is free entry and exit, those bankers who manage to attract depositors in equilibrium earn strictly higher returns than those who do not. The inactive bankers invest in the storage technology and cannot disrupt the positive-profit equilibrium. The usual indifference condition governing entry and exit breaks down because of the interaction between the informational constraints.

Preview of the model and results.

In the model, ex ante identical bankers, each with an initial endowment of one unit, compete to attract depositors. Depositors choose between delegating the management of their funds to a banker and investing in a storage technology. Bankers who attract sufficient depositors invest both their own endowment and the collected funds directly in a project; those who fail to attract depositors fall back on the storage technology. The model features two frictions: ex ante effort moral hazard and ex post diversion.

Banks can improve the probability of the project's success by exerting costly and unobservable effort (see, e.g., Holmstrom and Tirole 1997, Morrison and White 2005, Mehran and Thakor 2011, He and Krishnamurthy 2012). The cost of effort is assumed to be sufficiently high that a banker investing only their own funds would not find it worthwhile to exert effort. One can think of this fixed cost as the acquisition of information that is reusable across borrowers, such as macroeconomic or sector-specific knowledge. The presence of a fixed cost is common in most classical theories of financial intermediation (e.g., Diamond 1984, Boyd and Prescott 1986, Williamson 1986a, and Williamson 1987).¹ However, when the banker also manages external funds and earns rents from doing so, effort exertion becomes incentive compatible. Hence, unlike the standard models of effort moral hazard, this incentive constraint imposes a lower bound, rather than an upper bound, on bank leverage.

¹As noted by Williamson (1986b), some degree of scale economy is generally necessary for delegated intermediation to dominate bilateral trading. Consistent with this premise, a large empirical literature finds that larger banks tend to exhibit lower average costs and benefit from increasing returns to scale (e.g., Wheelock and Wilson 2012, Hughes and Mester 2013, Davies and Tracey 2014, and Kovner et al. 2014).

Banks also possess a diversion technology which allows the manager to divert a fraction of the final cash flow from depositors (see, e.g., Calomiris and Kahn 1991, Hart and Moore 1995, Shleifer and Wolfenzon 2002, Holmstrom and Tirole 2011, Gennaioli and Rossi 2013, Donaldson et al. 2018, and Biswas and Koufopoulos 2022). Calomiris and Haber (2014) present a compelling case for diversion to be considered a central friction in the banking industry. They highlight several channels through which bank insiders can extract value: politically connected bankers may extend loss-making loans to themselves and connected firms², pay themselves dividends even when the bank is distressed³, and shift risks onto depositors or taxpayers through implicit or explicit guarantees. In environments where corrupt politicians shield bank insiders from effective oversight, insiders can systematically divert resources from the institution and eventually walk away when insolvency becomes unavoidable.⁴

In the unique equilibrium of our model, both the diversion constraint and the effort incentive constraint bind. Suppose that the effort incentive constraint is just binding, but the diversion constraint is slack. Then, a bank can increase its profits by offering a slightly higher deposit rate and attracting more depositors. Conversely, suppose that the diversion constraint is just binding, but the effort incentive constraint is slack. Then, an inactive banker in this allocation can increase his profits by offering a slightly higher deposit rate to attract depositors from the incumbents. Finally, there is no profitable deviation from the intersection of the two constraints. At this allocation, an inactive banker cannot attract depositors away from incumbents by offering a higher deposit rate,

²See also the case of the failure of the Baoshang Bank in China.

³In the context of the 2007-2009 financial crisis, Acharya et al. (2011) document that banks continued to pay substantial dividends despite widely anticipated credit losses, thereby transferring value from creditors to shareholders. Goetz et al. (2025) find that banks with higher pre-crisis insider stakes issued significantly less common equity during the crisis period, arguably to avoid diluting insiders' private benefits of control.

⁴They highlight Mexico as a prominent example. Following the reprivatization of banks in the early 1990s in which wealthy families gained access to bank ownership through political connections, the bankers engaged in extensive insider lending and paid themselves overly generous dividends, leaving institutions undercapitalized leading up to the mid-90's Tequila Crisis. When losses materialized during the crisis, the government absorbed bad loans through FOBAPROA, effectively socializing the losses while insiders had already captured private gains. By contrast, they note that Canada's institutions have prevented such rent extraction by bank insiders, and as a result the country has experienced far fewer financial crises.

as such an offer would violate at least one of the two informational constraints.

In this equilibrium, banks make strictly positive profits even though there is costless free entry and price competition. This is because the interaction of the diversion constraint and the moral hazard constraint prevents undercutting beyond a point, allowing banks to offer a deposit rate which is lower than the competitive rate. As we clarify in Section 3.2, if either the diversion constraint or the moral hazard constraint is removed from the model, the outcome reverts to the Bertrand benchmark. Thus, it is the interaction of the two frictions, rather than either friction alone, that drives the result that banks possess pricing power. To see it more clearly, we consider the case that effort is contractible. We show that in this case the diversion constraint still binds, but active and inactive bankers earn the same expected return in equilibrium.

Empirical predictions and policy implications.

The interaction of the diversion constraint with the effort moral hazard constraint produces a number of new implications:

Consistent with the evidence in Smirlock (1985), Berger (1995), and Claessens and Laeven (2004), our model predicts that rising bank concentration can coincide with declining bank profitability. Jackson (1992) documents a non-linear relationship between concentration and pricing power in the deposit market, with a negative correlation emerging for banks operating in highly concentrated markets. Likewise, using a cross-country sample, Forssbaeck and Shehzad (2015) find that concentration and pricing power are negatively related in deposit markets. Our model explains this decoupling between concentration and pricing power. As pledgeability rises, banks offer higher deposit rates and operate with higher leverage, yet increased leverage leads to fewer banks in equilibrium (higher concentration). We refer to this result as the dichotomy of concentration and pricing power.

If investment opportunities improve, banks can lever up more since the total pledgeable in-

come rises, all else given. At the same time, there is a countervailing force: as investment opportunities improve, bankers earn higher rents per unit invested. As a result, they require fewer depositors to make effort exertion incentive compatible. We show that the second effect dominates leading to a decline in leverage when investment opportunities improve. The extent to which an increase in profitability due to stronger investment opportunities is passed through to depositors depends on pledgeability. The relationship between the strength of the pass-through effect and pledgeability is non-monotonic: with an increase in pledgeability, pass-through becomes more complete when pledgeability is low but less complete once pledgeability becomes sufficiently high. The strength of the pass-through effect is monotonically increasing in banks' investment opportunities. We present several other new predictions in Section 4.1.

In terms of policy implications, we show that the number of banks is excessively high in the unregulated equilibrium. A utilitarian regulator can step in to correct this inefficiency by imposing an upper bound on the number of banking licenses issued. In this regulated equilibrium, bankers who succeed in attracting depositors earn even higher returns relative to inactive bankers than they would in the unregulated case.

Related literature.

Although theories featuring switching costs can potentially explain the emergence of pricing power (e.g. Farrell and Klemperer 2007), they cannot predict the negative correlation between concentration and pricing power. Furthermore, we model the uninsured deposit market. This segment of the deposit market features large deposits. Drechsler et al. 2024 provide evidence that even though pricing power is somewhat smaller in the uninsured segment of the deposits market compared to insured segment, it remains substantial there. At the same time, fixed switching costs are less likely to be a significant factor in determining depositor behavior in this segment due to the size of the deposits. Finally, switching costs have become increasingly less relevant given recent

regulation.⁵ These observations suggest that existing models relying on switching costs are ill-suited for explaining pricing power in the large-deposit segment of the market. A new framework is therefore needed to rationalize substantial pricing power in the large-deposit market segment, while also accounting for the observed inverse relationship between market concentration and pricing power. Our model aims to fill this gap.

In one set of models, asymmetric information on the asset side endogenously restricts competition in the banking sector which enables banks to earn information rents (see, e.g., Broecker 1990, Shaffer 1998, Dell’Ariccia et al. 1999, Boot and Thakor 2000, Marquez 2002, and Sengupta 2007). In contrast, we focus on the funding side of banks. As noted by Beyhaghi et al. (2025), the nature of informational frictions on the asset side differs from those on the funding side. For example, a key feature of the asset-side models is that some banks possess superior information about certain borrowers’ credit quality. This type of friction is not relevant in the deposit market; hence, we consider a distinct problem. Another key difference is conceptual. In much of the existing literature, the structure of the banking industry is imposed rather than derived; even when entry is allowed, the baseline market configuration is typically taken as given (e.g., by assuming N incumbent banks that are endowed with some form of informational advantage relative to potential entrants). By contrast, we do not impose any default structure. Instead, we endogenize the emergence of market structure as an equilibrium outcome, which allows us to generate new insights.

Our paper is also related to models of bank competition and financial stability (see, e.g., Hellmann et al. 2000, Repullo 2004, Martinez-Miera and Repullo 2010, and Vives and Ye 2025). As in the asset-side literature, these models typically take the baseline market configuration as given. In contrast, we endogenize the market structure as part of the equilibrium. Furthermore, while

⁵E.g., in Europe, the open banking provisions in the Payment Services Directive 2 shifts the burden of switching from depositors to banks.

much of the existing literature assumes some form of product differentiation (imperfect competition), we allow for price competition for a homogeneous product, reflecting the idea that deposit contracts across banks offer similar, if not identical, services to depositors. As with the switching cost models, these models cannot explain the negative correlation between concentration and pricing power. Finally, once free entry is permitted in these models, the equilibrium number of banks goes to infinity, driving the price down to the Bertrand level.

Several models of imperfect competition in banking study how monetary policy shocks affect deposit franchise values and their transmission to the real economy (e.g., Drechsler et al. 2017, Drechsler et al. 2021, Wang et al. 2022, Drechsler et al. 2024). Our model introduces a channel through which shocks to banks' investment opportunities is transmitted to the liability side. Specifically, a positive shock to investment opportunities results in higher deposit rates, but leverage declines. By endogenizing the source of banks' pricing power our approach offers richer predictions. These models typically interpret the partial pass-through of shocks to depositors as evidence of market power arising from switching costs or imperfect competition. In our model, however, pass-through can be partial even when there is free entry and exit and banks compete in prices. We also generate new predictions regarding the nature of the pass-through effect.

In our model, both pricing power and market concentration are determined endogenously in equilibrium, thereby making any correlation a spurious one. The efficiency structure hypothesis (see, e.g., Demsetz 1973, Demsetz 1974, and Peltzman 1977) similarly argues that more efficient firms expand their market share, leading to greater market concentration, and that their lower costs translate into higher profits leading to a spurious correlation between concentration and pricing power. Relatedly, Lambson (1987) shows that lumpy or indivisible technologies can generate a spurious positive correlation between concentration and profitability. Unlike our framework, however, these models do not allow for price competition and free entry, and rely on assumptions

such as asymmetric technologies to microfound positive profits or adopt Cournot competition structures.

Our work is also related to static models in industrial organization where price competition does not necessarily lead to zero profits which is the standard Bertrand result. Kreps and Scheinkman (1983) analyze a two-stage game where firms first commit to a production capacity which softens price competition in the subsequent stage (see Marquez 2002, Schliephake and Kirsten 2013, and Green 2007 for applications in banking and finance).

In our model, even though leverage (quantity) and the deposit rate (price) are determined simultaneously, as opposed to quantity first and price second as in the Kreps and Scheinkman (1983) game, price competition does not lead to the Bertrand outcome. Moreover, as we show in the Appendix (Section 6.2), if we adapt the Kreps and Scheinkman (1983) game and allow for costless free entry by banks at the ex ante stage (as we do in our model), then the unique equilibrium is the Bertrand outcome. This highlights that, despite the similarities at a superficial level, the constraints on competition in the two models work fundamentally differently. Finally, Kreps and Scheinkman (1983) consider a pre-specified and fixed number of firms without commenting on how the market structure comes to be, whereas, in our model, the number of banks in equilibrium is endogenous.

Maggi (1996) observe that in the Kreps and Scheinkman (1983) world, the equilibrium outcome is Bertrand in the absence of capacity constraints, and Cournot when such constraints are present. By allowing firms to produce quantities beyond the announced capacities at a higher marginal cost, Maggi (1996) captures intermediate situations between these two extremes. However, this result also relies on the assumption of fixed number of firms in the industry, since if we allow for free entry in this model, the unique equilibrium will be the Bertrand outcome (following the same logic that applies to the Kreps and Scheinkman 1983 model). In our model, despite free

and costless entry and exit the markup lies within a range from monopoly to perfect competition and it is determined by the deep parameters underlying the diversion and effort moral hazard constraints.

2 Model

We develop a model in which banks raise funds in the deposit market and invest them directly. Direct investment by banks may be thought of as fee-generating investment banking activities. Another equivalent interpretation is that banks behave monopolistically in the loan market and extract the full surplus in this market. We show that, even though entry is costless, banks make strictly positive profits in equilibrium.

2.1 Set-up

We consider an economy with two types of agents: a measure $m \gg 1$ of bankers who may raise funds from external investors, and a separate set of agents of measure 1 who may delegate the management of their funds to a banker (depositors) or invest in an unproductive storage technology (which produces a zero net return). Each agent is endowed with an initial wealth of 1 unit.

There are two dates, $t \in \{0, 1\}$. At $t = 0$, bankers offer contracts to depositors. Entry into banking is costless. A fraction $\lambda \in [0, 1]$ of bankers raise sufficient funds to operate a bank, where λ is determined in equilibrium by agents' optimizing behavior. There are $n = \lambda m$ bankers who successfully raise funds from depositors and invest them in a project. Those bankers who fail to do so invest their funds in the unproductive storage technology. At $t = 1$, project returns are realized. Per unit of investment, the project yields X if it succeeds and 0 otherwise. The project exhibits constant returns to scale and is infinitely scalable.

There are two informational frictions:

Effort moral hazard. After obtaining the funds through deposits, a banker can choose to exert effort or not. The project succeeds with probability p_h and fails with probability $1 - p_h$ if the banker exerts effort. Effort is costly (in terms of utility) and consists of two components: a fixed cost, k , and a variable cost, c , per unit of investment. The fixed cost is interpreted as the acquisition of information that is reusable across borrowers (such as macroeconomic or sector-specific knowledge), while the variable cost reflects the increasing difficulty of managing a larger bank; both components are incurred *after* a banker has collected deposits and the scale of the bank is known. If the banker does not exert effort, the project succeeds with a lower probability, $p_l < p_h$. We normalize the return from a project when the banker shirks to equal the return from the storage technology, i.e., $p_l X = 1$. We denote the incremental success probability from exerting effort, $p_h - p_l$, as Δ . Effort exertion is not observable or verifiable, and hence, effort is not contractible.

Diversion. A banker can divert a fraction, $(1 - \phi)$, of the realized output at $t = 1$, $\phi \in (0, 1)$. Diversion of funds is not verifiable, and hence, the banker cannot credibly commit ex ante to not diverting these funds even if he would like to be able to do so, i.e., pledgeability is limited.

We make the following parametric assumptions:

$$A1: k > \Delta X - c > \phi \Delta X$$

$$A2: \phi < \frac{1}{p_h X}$$

$$A3: m > \frac{(1-\phi)\Delta X - c}{k + c - (1-\phi)\Delta X}$$

The first inequality in Assumption A1 says that the fixed cost of effort is large implying that exerting effort would be worthwhile only if multiple agents pool their resources and share the cost (see also, Boyd and Prescott 1986); thus, unless a banker raises sufficient funds in the deposit market, he invests his endowment of one unit in the storage technology. The second inequality in Assumption A1 says that the marginal cost of effort is small. Assumption A2 says that diversion

is sufficiently large. These assumptions ensure that the two information constraints are binding in equilibrium. Assumption A3 implies that, given binding informational frictions, there are sufficiently many potential bankers to serve all depositors, so some bankers fail to attract external funds.

2.2 Equilibrium

A contract is a triple (d, R, I) , where d is the number of deposits the bank accepts, R is the deposit rate, and $I \in \{0, 1\}$ is an indicator variable which equals 1 if the banker commits to contributing his own endowment to the project and 0 otherwise.

The following lemma pins down I and simplifies the subsequent analysis.

Lemma 1 *The contract sets $I = 1$: a banker commits to contributing his own endowment to the bank he manages.*

Proof. The proof is in the Appendix. ■

Lemma 1 establishes that the banker always invests his own endowment alongside externally raised deposits; if the banker does not commit to doing so, depositors would not recover their funds in expectation and will never choose such a contract. A bank's total investment is therefore $(1 + d)$ units. With probability p_h or p_l (depending upon the level of effort exertion), these funds produce X (per unit), and they produce 0 with the complementary probability. In the event of success, the banker repays dR to depositors. A feasible contract must satisfy four constraints:

1. the diversion constraint: If the project fails, there are no cash flows to disburse and nothing to divert. In the case that the project succeeds, due to the possibility of diversion, the total verifiable cash flow is $\phi(1 + d)X$. This total verifiable cash flow must be weakly greater than

the promised return to the depositors, dR .

$$\phi(1 + d)X \geq dR \quad (1)$$

2. the effort incentive constraint: exerting effort leads to a higher success probability, but it imposes a cost on the banker. A banker will exert effort only if he is (weakly) better off from exerting effort than not. The incentive constraint is written as follows:

$$\begin{aligned} (1 + d)p_h X - dp_h R - 1 - k - (1 + d)c &\geq (1 + d)p_l X - dp_l R - 1 \\ \implies d[\Delta(X - R) - c] &\geq k + c - \Delta X \end{aligned} \quad (2)$$

3. the depositors' participation constraint: the depositors must be weakly better off from depositing their funds with the bank, in which case they obtain, $p_h R$, than investing in the storage technology.

$$p_h R \geq p_l X = 1 \quad (3)$$

4. the banker's participation constraint: It must be the case that managing a bank is (weakly) more profitable than investing in the storage technology:

$$\Pi = (1 + d)p_h X - dp_h R - 1 - k - (1 + d)c \geq 0 \quad (4)$$

We show below that this constraint is always slack.

We consider each of the constraints in turn:

Diversion constraint. First, we re-write the diversion constraint as follows:

$$d \leq \frac{\phi X}{R - \phi X} \quad (5)$$

The diversion constraint puts an upper bound on the amount of funds that the depositors are willing to provide to the bank. To derive the slope of the diversion constraint in the (d, R) space, we totally differentiate Equation (5) with respect to d and R :

$$\left. \frac{dd}{dR} \right|_{DC} = -\frac{d}{R - \phi X} < 0 \quad (6)$$

The slope of the diversion constraint is negative (and it is convex). This reflects that as each depositor is promised a higher repayment, the bank can accept fewer deposits since the diversion constraint tightens. A feasible equilibrium must lie on or below the diversion constraint.

Depositors' PC. The depositors' participation constraint, $R = \frac{1}{p_h} = R^P$ is a vertical line in the (d, R) space. A feasible equilibrium must lie to the right of the depositors' participation constraint.

Bankers' IC. Next we turn to the incentive constraint. In contrast to existing models featuring this friction (e.g., Holmstrom and Tirole 1997), it imposes a *lower* bound on bank leverage. The intuition is as follows. Given Assumption A1 that the fixed cost of effort is high, $k > \Delta X - c$, it is not worthwhile for bankers to exert effort when investing solely their own funds. Effort becomes worthwhile only when the banker manages external funds and thereby obtains rents from doing so. For any R , if a given level of deposits d provides sufficient rents for the banker to satisfy the incentive constraint, then the constraint must also hold for a higher d .

To derive the slope of the incentive constraint in the (d, R) space, we totally differentiate Equa-

tion (2) with respect to d and R :

$$\left. \frac{dd}{dR} \right|_{IC} = \frac{\Delta d}{\Delta(X - R) - c} > 0 \quad (7)$$

The effort incentive constraint has a positive and convex slope. The positive slope reflects that as depositors are offered a higher promised repayment, bankers retain a smaller share of the surplus, thereby reducing their incentive to exert effort. Starting from a (d, R) pair that lies on the incentive constraint – such that the bank is just willing to exert effort – an increase in R must be accompanied by a higher d in order to preserve incentive compatibility. Moreover, as R increases, the slope of the constraint becomes steeper, approaching ∞ as $R \rightarrow X - \frac{c}{\Delta}$ (i.e., the constraint becomes vertical in the limit). A feasible equilibrium must lie on or above the incentive constraint.

Bankers' PC. Finally, we consider the banker's participation constraint.

Lemma 2 *For any $R < X$, whenever the effort incentive constraint of the banker is satisfied, his participation constraint is slack.*

Proof. The incentive constraint requires $R \leq X - \frac{c}{\Delta}$, so any incentive-compatible contract necessarily satisfies $R < X$. The left-hand side of the incentive constraint coincides with that of the participation constraint, while the right-hand side of the incentive constraint is strictly positive for $R < X$. Since the participation constraint has a right-hand side equal to zero, it follows that whenever the incentive constraint holds, the participation constraint is automatically satisfied with slack. ■

Equilibrium. Having introduced the four constraints, we are now in a position to define the equilibrium concept. We adopt a *free-entry competitive equilibrium*: a contract is taken as given by individual agents, who choose whether to participate; the deposit market clears through the

endogenous adjustment of the measure of active banks; and the equilibrium is robust to entry by inactive bankers, in the sense that no inactive banker can attract depositors away from incumbents and earn strictly positive profits by offering a feasible alternative contract.

Definition 1 *A free-entry competitive equilibrium is a triple $(R^*, d^*, \lambda^*) \in \mathbb{R}_{++} \times \mathbb{R}_{++} \times [0, 1]$ together with an effort choice $e^* = 1$, such that:*

(E1) **Feasibility.** *The contract (R^*, d^*) satisfies the diversion constraint, the effort incentive constraint, the depositors' participation constraint, and the bankers' participation constraint; each active banker invests his own endowment, $I^* = 1$ exerts effort, $e^* = 1$.*

(E2) **Market clearing.** $\lambda^* m d^* = 1$.

(E3) **No profitable entry.** *There exists no alternative contract (R', d') satisfying the diversion and incentive constraints such that $R' > R^*$ and $\Pi(d', R') > 0$.*

Conditions (E1) states that a feasible contract must respect the informational frictions, and both bankers and depositors must (weakly) prefer participation to their outside options (storage). (E2) is the market clearing condition: the aggregate demand for deposits is endogenous – both the measure of active banks, $\lambda^* m$, and the deposits per bank, d^* , are determined in equilibrium – while aggregate supply is fixed at the unit measure of depositors. (E3) is a free-entry condition: it captures the Bertrand-style undercutting that price competition would induce. The equilibrium contract must be such that no inactive banker can profitably attract depositors by offering a strictly higher deposit rate.⁶

Increasing the fixed cost of exerting effort, k , makes the effort incentive constraint tighter, but it does not affect any of the other constraints. The intersection of the effort incentive constraint and

⁶The strict inequality $R' > R^*$ in (E3) can be justified by an arbitrarily small switching cost $\varepsilon > 0$ borne by depositors: under this tie-breaking rule, depositors switch from an incumbent's contract to an entrant's contract only if the entrant's rate strictly exceeds the incumbent's. The same tie-breaking rule is used implicitly in (E1): indifference between depositing and storage is resolved in favor of depositing, and indifference between two contracts with the same rate is resolved in favor of the incumbent.

the diversion constraint depends on the fixed cost of effort. The lower the cost, the intersection of the two constraints is further to the right in the (d, R) space. Given the parametric restrictions made above, there are two cases to consider:

1. $k > \frac{(1-\phi)\Delta X - c}{1-\phi p_H X} \equiv \bar{k}$
2. $\bar{k} \geq k > \Delta X - c$

2.2.1 Case 1: $k > \bar{k}$

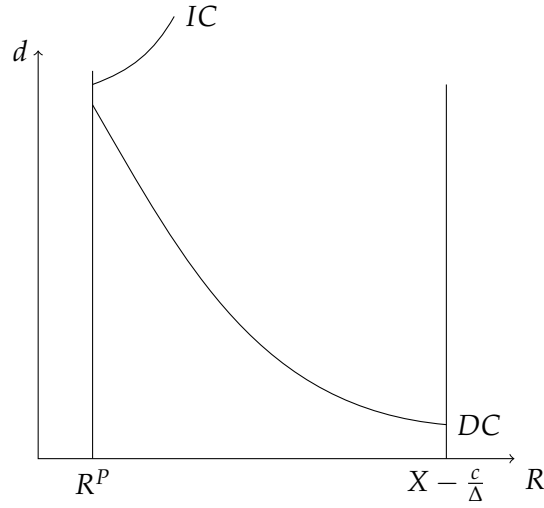


Figure 1: No banks

We illustrate the depositors' participation constraint, the diversion constraint, and for $k > \bar{k}$, the bankers' incentive constraint in Figure 1. In this case, the incentive constraint intersects the diversion constraint to the left of the depositors' participation constraint. Bankers exert effort for a (d, R) pair which lies above the incentive constraint. It can be seen that the set of parameters where bankers exert effort, and at the same time, none of the constraints is violated, is empty. That is, the cost of effort is so high that the surplus generated from effort exertion is always strictly less than the cost. Anticipating that it is never incentive-compatible for banks to exert effort – and thus that depositors would not deposit their funds in banks – none of the bankers make an offer to attract

depositors. Instead, all agents (both bankers and depositors) invest in the storage technology. We summarize this results in the proposition below:

Proposition 1 (No banks) *Suppose that $k > \bar{k}$. There are no banks and all agents invest in the storage technology.*

2.2.2 Case 2: $\bar{k} \geq k > \Delta X - c$

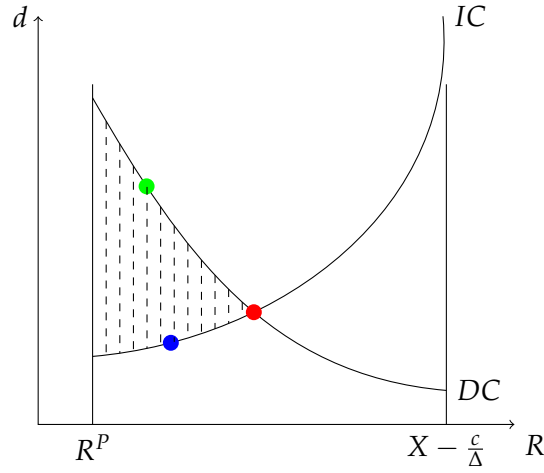


Figure 2: Pricing power

In Figure 2, we consider the $\bar{k} > k > \Delta X - c$ case. In this case, the incentive constraint intersects the diversion constraint to the right of the depositors' participation constraint and this intersection is represented by the red dot. If banks do not exert effort, their investment will be negative-NPV, and depositors will not be able to recover their funds. Hence, banks will not obtain funds, implying that any (d, R) pair below the IC constraint can be ignored. By a similar argument, contracts above the diversion constraint are infeasible, since depositors would not be fully repaid. Thus, the feasible contracts are in the shaded area. We argue below that the unique free-entry competitive equilibrium is at the intersection of the two constraints (the red dot). Using the diversion constraint and the incentive constraint, we derive the expressions for d and R at the

red dot:

$$d = \frac{k + c - (1 - \phi)\Delta X}{(1 - \phi)\Delta X - c} \equiv d^* \quad (8)$$

$$R = \frac{\phi k X}{k + c - (1 - \phi)\Delta X} \equiv R^* \quad (9)$$

Given Assumptions A1 and A2, $d^* > 0$ and $R^P \leq R^* < X$. Our main result is that at the red dot, bankers who successfully attract depositors (active bankers) make strictly higher expected profits than inactive bankers despite ex ante identical bankers, free entry, and price competition.

We present this result in the proposition below:

Proposition 2 (Pricing power) *Suppose that $\bar{k} \geq k > \Delta X - c$. A fraction λ^* of bankers manage banks and all depositors are served. The unique equilibrium is characterized by (d^*, R^*, λ^*) , where:*

$$\lambda = \frac{(1 - \phi)\Delta X - c}{m[k + c - (1 - \phi)\Delta X]} \equiv \lambda^* \quad (10)$$

Proof. The proof is in the Appendix. ■

The unique equilibrium is at the red dot. By Definition 1, no banker can profitably deviate: a more attractive offer ($R = R^* + \epsilon$) violates at least one of the two informational constraints and is therefore infeasible, while inactive bankers cannot attract depositors away from incumbents under the tie-breaking rule. To derive the equilibrium fraction of active bankers, λ^* , we use the market clearing condition (E2), $m\lambda^*d^* = 1$. Under the parametric assumptions, $\lambda^* < 1$, so a positive mass of bankers remain inactive in equilibrium and invest in the storage technology.

Pricing power. We refer to this as the equilibrium with pricing power. As shown in Lemma 2, the bankers' participation constraint is slack whenever the incentive constraint binds, which is the case at the red dot. That is, banks offer a deposit rate low enough to earn strictly positive expected

profits at this allocation. Despite these profits and costless free entry, a strictly positive mass of inactive bankers remain, investing in the storage technology. They cannot disrupt the equilibrium due to the binding informational constraints. Thus, although bankers are identical to begin with, and inactive bankers may offer contracts at zero cost in an attempt to undercut incumbents, active bankers earn strictly higher profits than inactive ones.

3 Discussion

3.1 Degree of competition

Monopoly pricing. Consider the special case, $k = \bar{k}$. In this case, the incentive constraint and the diversion constraint intersect on the depositors' participation constraint, which is characterized as follows:

$$d = \frac{\phi p_h X}{1 - \phi p_h X} \equiv d^M \quad (11)$$

$$R = \frac{1}{p_h} = R^P \quad (12)$$

In this equilibrium, the depositors are on their participation constraint. Hence, we refer to this as the monopoly pricing equilibrium.

Perfect competition. Suppose that $k = \Delta X - c + \epsilon$, where ϵ is strictly positive but arbitrarily small. Substituting into Equation (9), as $\epsilon \rightarrow 0$, we obtain:

$$R^* \rightarrow X - \frac{c}{\Delta} \quad (13)$$

This implies that as k becomes very small (while still satisfying Assumptions A1-A2), the equilibrium is the same as the Bertrand outcome under only effort moral hazard (in absence of the diversion constraint) in which the margin that banks charge is arbitrarily close to the gross informational rent, $\frac{c}{\Delta}$.

Thus, the level of competition, whether it is monopoly, oligopoly, or perfect competition, is traced to the primitives of the economy, as summarized in the following proposition:

Proposition 3 *The degree of competition lies within a range from monopoly to perfect competition and is determined by the deep parameters:*

1. $k = \bar{k}$: Monopoly pricing, $R = R^P$.
2. $\bar{k} > k > \Delta X - c$: Oligopoly, $R = R^* > X - \frac{c}{\Delta}$.
3. $k \rightarrow \Delta X - c$: Perfect competition, $R \rightarrow X - \frac{c}{\Delta}$.

This result is reminiscent of Maggi (1996) who captures all situations between the Cournot and the Bertrand outcome, depending on parameters. Different from Maggi (1996), the number of banks in our model is endogenous as we allow for free entry and exit. As in the Kreps and Scheinkman (1983) model, when we allow for free costless entry in Maggi (1996), the unique equilibrium is the Bertrand outcome.

3.2 Both frictions matter

Strikingly, even though entry and exit are costless, banks make strictly positive profits in equilibrium. Indeed, these profits exceed the informational rent, $\frac{c}{\Delta} - c$. At first sight, it may seem that this result is driven by the diversion constraint which acts as a capacity constraint. However, as we show below, if either the diversion constraint or the effort moral hazard constraint is removed,

the outcome reverts to the corresponding Bertrand benchmark. Thus, it is the interaction of the two frictions, rather than either friction alone, that drives the results.

To see it clearly, consider the case in which effort is costless, $k = c = 0$ (violating parametric restrictions). In this case, the equilibrium deposit rate will be equal to that under full information, i.e., $R = X$. Hence, given free entry and exit, bank profits will be 0 if the only constraint faced by banks is the diversion constraint. As before, the equilibrium must lie on the diversion constraint. Suppose the equilibrium occurs at some $R < X$. Then, bankers make strictly positive profits. Since entry is costless, an inactive banker in this allocation can profitably deviate by offering a deposit contract with a slightly higher deposit rate, attracting depositors away from incumbent banks. This process iterates until $R = X$. Thus, when effort is costless and the only relevant friction is limited pledgeability, the Bertrand outcome obtains.

Lemma 3 (Costless effort) *If $k = c = 0$, the Bertrand outcome obtains in which case, $R = X$.*

Thus, Lemma 3 implies that while the diversion constraint in our model resembles a capacity constraint in the spirit of Kreps and Scheinkman 1983, it does not, by itself, yield strictly positive profits under free entry.

Next, consider the case that pledgeability is perfect, $\phi = 1$ (violating parametric restrictions). In this case, the diversion constraint no longer binds, so there is no upper limit on the amount of deposits a bank can accept. For any $R < X - \frac{c}{\Delta}$, banks would undercut each other by offering a slightly higher deposit rate (by an arbitrarily small ϵ) to capture the entire market, implying that the Bertrand logic applies. Any offer above $X - \frac{c}{\Delta}$, however, would violate the incentive constraint and is therefore infeasible.

Lemma 4 (No diversion) *If $\phi = 1$, in equilibrium, a bank attracts the full measure of depositors by offering $R \rightarrow X - \frac{c}{\Delta}$.*

In Kreps and Scheinkman (1983), a capacity constraint along with an exogenously fixed number of firms in the industry chokes competition and allows these firms to make strictly positive profits. As we show in Section 6.2, when we allow for costless free entry in the Kreps and Scheinkman (1983) game, the Bertrand outcome obtains as the unique equilibrium. In contrast, in our model, the positive profits arise due to the interaction of two informational constraints (neither of which can be contracted away) even under free costless entry and exit in the industry. This contrast highlights that, despite superficial similarities, the mechanisms at work are fundamentally different.

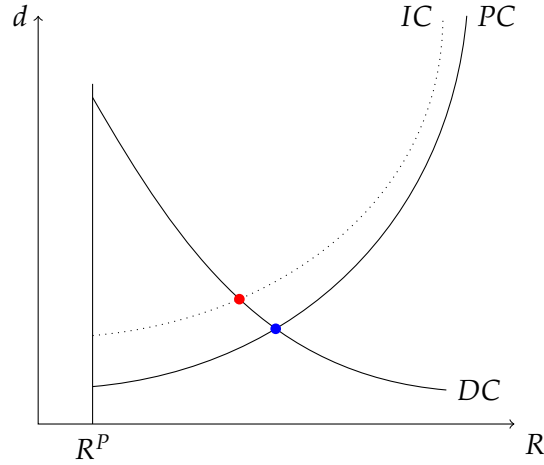


Figure 3: Contractible effort

Finally, consider the case that effort is costly but contractible. In this case, the IC constraint is no longer relevant (the dotted curve in Figure 3), since contracts can specify not only R but also the level of effort. Same as the IC constraint, the banker's participation constraint (Equation (4)) is upwards sloping in the (d, R) space and it becomes vertical as R approaches $X - \frac{c}{p_h}$. As argued in Lemma 2, for any $R < X$, the banker's participation constraint lies strictly below the IC constraint (see Figure 3). Contractibility of effort enlarges the set of feasible equilibria to also include the region between the IC constraint and the banker's participation constraint.

As before, the diversion constraint must bind. The unique equilibrium is given by the inter-

section of the diversion constraint and the banker's participation constraint (the blue dot in Figure 3). No agent can profitably deviate from this allocation since a banker will be strictly worse off if he attempts to undercut another and move further down along the participation constraint. At this allocation, active bankers and those who do not attract depositors earn identical expected returns because the banker's participation constraint binds. Therefore, the result that Bertrand competition is disrupted despite costless free entry and price competition relies crucially on the assumption of non-contractibility of effort.

Lemma 5 (No moral hazard) *Under contractible effort, bankers who successfully attract depositors and those who do not earn identical expected returns.*

We summarize these results in the proposition below:

Proposition 4 *The interaction of effort moral hazard and diversion frictions, not either friction alone, prevents the equilibrium bank profits from being driven down to zero, the Bertrand outcome.*

4 Implications

In this section, we present the empirical and policy implications of the model which arise from the interaction of the diversion constraint and the ex ante effort moral hazard constraint.

4.1 Comparative statics

In this section, we derive some novel empirical predictions out of the model. Consider the case that the cost of effort is in the intermediate range, $0 < k < \bar{k}$, implying that banks retain a part of the surplus.

Lemma 6 *An increase in banks' investment opportunities, X , leads to more banks, λ^* , a higher deposit rate, R , and a lower bank leverage, d .*

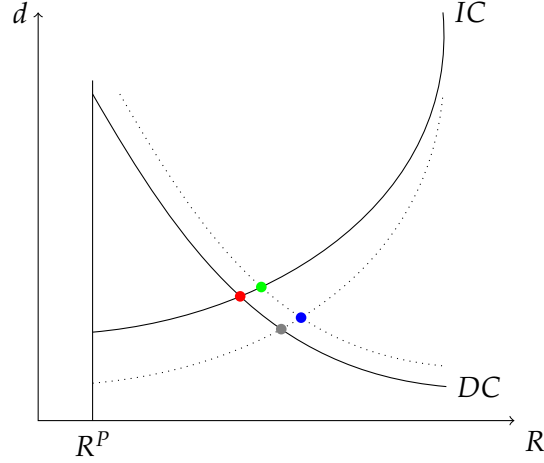


Figure 4: Comparative Statics

Proof. The proof is in the Appendix. ■

When the bank's investment opportunities improve, which is captured by a higher X , there are two forces at play:

It relaxes both the diversion constraint and the effort incentive constraint. The new intersection of these two constraints is at the blue dot. Both forces have a positive effect on the deposit rate, which increases as a result of a higher X . Interestingly, the two forces have conflicting effects on leverage – the first effect boosts leverage, taking as given the number of banks operating, but the second effect boosts the number of banks operating in equilibrium (λ increases), which leads to a lower bank leverage. We show in the proof of Lemma 4 that the second effect dominates, and the new equilibrium features a lower d when X increases – in Figure 4, the equilibrium moves from the red to the blue dot. That is, even though banks' investment opportunities improve allowing banks to attract more deposits (all else given), more bankers are active in equilibrium due to the relaxation of the effort incentive constraint, and this leads to a lower equilibrium leverage.

Lemma 7 *An increase in pledgeability, ϕ , leads to fewer banks, a higher deposit rate, R , and a higher bank leverage, d .*

Proof. The proof is in the Appendix. ■

When pledgeable income increases because diversion, $1 - \phi$, falls, the diversion constraint is relaxed while the other constraints are unaffected. The equilibrium moves up along the incentive constraint implying that there are fewer banks, and it features a higher R and a higher d – in Figure 4, the equilibrium moves from the red to the green dot. Intuitively, a higher ϕ implies that banks can pledge more of their output to depositors. This intensifies competition for depositors leading to higher deposit rates. As a result, per unit of investment, banks retain a smaller surplus, and an increase in leverage is necessary to satisfy the effort incentive constraint.

Various regulatory measures over the years have fortified stability in the banking system and presumably led to greater pledgeability. Our model thus predicts the banking sector should get more concentrated over time featuring fewer banks and banks should be more levered. There are numerous studies providing supporting evidence for these predictions: Saunders and Wilson (1999) document that the banking sectors in the US, UK, and Canada have become increasingly consolidated over the very long run, and at the same time, banks have become more levered. Corbae and D’Erasmus (2020) document increasing concentration in the US banking sector and Miles et al. (2013) find that UK banks have become more levered over time.

If we assume that the strength of institutions is increasing in GDP/capita, then richer countries are characterized by a lower threat of diversion. Given this, the prediction is that bank pricing power is falling in GDP/capita, but the banking sector is more concentrated (fewer banks).

Thus, when pledgeable income increases, the effect on the deposit rate is unambiguously positive regardless of whether X or ϕ contributes to the increase in pledgeable income – higher pledgeable income leads to lower pricing power in the deposit market. However, the effect of a higher pledgeable income on bank leverage may be positive or negative depending upon whether ϕ or

X contributes to the increase in pledgeable income. If pledgeable income increases due to institutional factors which allows banks to commit to diverting a smaller fraction of the output (e.g., the regulatory environment), then it leads to higher leverage. However, if pledgeable income increases due to improved investment opportunities, then it leads to lower leverage.

Lemma 8 *A fall in the cost of effort, c or k , leads to more banks, a higher deposit rate, R , and a lower bank leverage, d .*

Proof. The proof is in the Appendix. ■

When the cost of effort, c or k , falls, it relaxes the effort incentive constraint while the other constraints are unaffected. The equilibrium moves down along the diversion constraint and features a higher R and a lower d – in Figure 4, the equilibrium moves from the red to the gray dot. This prediction aligns with the intuition that lowering the cost of effort increases the number of banks. Somewhat less obvious, however, is the accompanying decline in leverage.

4.2 The dichotomy of concentration and pricing power

High concentration in the banking sector can coexist with low pricing power.

When pledgeability increases, banks can offer higher deposit rates and accept more deposits. This, in turn, leads to fewer banks. To see this, note that a higher ϕ relaxes the diversion constraint while leaving the other constraints unaffected. As shown in Figure 4, the equilibrium shifts from the red to the green dot. This new equilibrium is characterized by fewer banks, and yet, a higher deposit rate. Since we assume that banks behave monopolistically in the loan market, a higher deposit rate in our model indicates lower profitability (controlling for bank size). Thus, a negative

correlation between pricing power in the deposit market, and hence, overall bank profitability, and bank concentration can emerge.

Consistent with this prediction, Smirlock (1985), Berger (1995), and Claessens and Laeven (2004) document that bank concentration and profitability can be negatively correlated. Likewise, Demirguc-Kunt et al. (2004) show that the positive correlation between concentration and net interest margins disappears once regulatory impediments to competition and inflation are taken into account. A small number of studies also directly look at the impact of concentration on pricing power in the deposit market. Jackson (1992) documents a non-linear relationship between concentration and pricing power in the deposit market, with a negative correlation emerging for banks operating in highly concentrated markets. Examining rural U.S. counties (those outside Metropolitan Statistical Areas), Rosen (2003) find that higher deposit market concentration can be associated with higher deposit rates. In an international sample of banks, Forssbaeck and Shehzad (2015) report a negative correlation between concentration and pricing power (Lerner index) in the deposit market, even though the correlation is positive in the loan market.

At the same time, holding the degree of diversion constant, an increase in the fixed cost of effort, k , makes the IC constraint more binding and reduces the number of banks, leading to higher concentration, and banks enjoy greater pricing power in this new equilibrium. Thus, our model also delivers the conventional prediction of a positive correlation between pricing power and concentration (see, e.g., Berger and Hannan 1989 for supporting empirical evidence).

Thus, there are countervailing forces which determine the correlation between concentration in the banking sector and the deposit rate, which can go either way depending on which of the two forces dominate. In general, although the US banking sector has become increasingly concentrated over the last few decades (as documented widely, see e.g., Corbae and D'Erasmus 2020), bank profitability has remained at a roughly similar level over time (see Drechsler et al. 2021), thereby

highlighting the disconnect between concentration and profitability. In our model, a simultaneous fall in diversion and an increase in the cost of effort (both, possibly due to heightened regulatory burden) can reconcile the observed rise in market concentration with stable profitability. Our analysis shows that treating concentration as an endogenous outcome rather than an exogenous variable yields a richer set of predictions.

4.3 The pass-through effect

The strength of the pass-through effect increases with pledgeability, ϕ , when ϕ is low, and decreases in ϕ when ϕ is high.

The extent to which an increase in X (e.g., due to a rise in the monetary policy rate) is passed through to depositors is given by:

$$\frac{\delta R^*}{\delta X} = \frac{\phi k(k+c)}{[k+c-(1-\phi)\Delta X]^2} \quad (14)$$

The pass-through effect is *partial* (i.e., the change in the equilibrium deposit rate is smaller than the change in X) if this derivative is less than one. Thus, there can be partial pass-through even when banks compete in prices, and there is neither product differentiation nor a switching cost for depositors.

The pass-through depends on pledgeability, ϕ :

$$\frac{\delta^2 R^*}{\delta X \delta \phi} = \frac{k(k+c)[k+c-(1+\phi)\Delta X]}{[k+c-(1-\phi)\Delta X]^3} \quad (15)$$

The pass-through is increasing in ϕ if:

$$\phi < \frac{k + c - \Delta X}{\Delta X} \quad (16)$$

This implies that when pledgeability is low, an increase in ϕ leads to greater pass-through, but when pledgeability is already high, the relationship may reverse. One could test this prediction using cross-country data. For example, an increase in pledgeability in Mexico (which is characterized by high diversion according to Calomiris and Haber 2014) should strengthen the pass-through effect, whereas the same increase in Canada or the United States should weaken it.

The pass-through also depends on investment opportunities, X . Specifically:

$$\frac{\delta^2 R^*}{\delta X^2} = \frac{2\phi k(k + c)(1 - \phi)\Delta}{[k + c - (1 - \phi)\Delta X]^3} > 0 \quad (17)$$

Since the second derivative is positive, the pass-through is convex in X . Thus, an increase in X has a larger effect on pass-through when X is already high, implying that pass-through becomes more complete at higher levels of X .

Lemma 9 *The strength of the pass-through effect is non-linear in ϕ and increasing in X .*

4.4 Optimal policy

A utilitarian regulator can eliminate a problem of excessively high number of banks by introducing an upper bound on the number of bank licences to be allocated.

When $k = \bar{k}$, there is monopoly pricing and the minimum number of bankers enter the market, $\lambda = 1 - \phi p_h X$, such that the market for deposits clears. Thus, in this monopoly equilibrium, the entire initial endowment of the economy is managed by banks, and at the same time, this

arrangement minimizes the exertion of costly effort. For $k < \bar{k}$, there is inefficient duplication of the fixed component of effort since there are too many banks, $\lambda > 1 - \phi p_h X$. As the fixed cost of effort, k falls, the net surplus in the economy increases, and more bankers are active in equilibrium. However, *for a given k* , net social surplus in the economy is maximized by minimizing the number of bankers to the extent that all depositors are served.

Suppose $k < \bar{k}$. A regulator whose objective is to maximize the net social surplus can eliminate the inefficiency by announcing a cap on the number of banking licences. The regulator will choose a maximum of $\frac{1-\phi p_h X}{\phi p_h X}$ (derived by setting $k = \bar{k}$ in λ^*) bankers to apply for a licence on a first-come-first-served basis. Since $k < \bar{k}$, the incentive constraint is slack and all chosen bankers exert effort. Thus, the problem of too many banks is eliminated since there are just enough banks in equilibrium such that all depositors are served. This regulated equilibrium resembles the unregulated monopoly equilibrium which arises when $k = \bar{k}$ in terms of the outcome variables, d and R . Compared to the laissez faire monopoly (i.e., when $k = \bar{k}$), active bankers in the regulated monopoly extract a higher fraction of the surplus.

Unlike the charter value hypothesis (see, e.g., Hellmann et al. 2000) in which a higher market power promotes stability by making banks more prudent in anticipation of higher charter values, in our theory, restricting competition in the banking sector improves efficiency by eliminating the unnecessary duplication of costly effort.

5 Conclusion

We develop a model of bank competition in the deposit market in which there are two informational frictions at play – an ex ante effort moral hazard problem and an ex post threat of diversion. The interaction of these two constraints (which cannot be contracted away) allows active banks to make strictly positive profits in the deposit market despite allowing for price competition and free

entry and exit; a strictly positive mass of inactive bankers who invest in the storage technology cannot disrupt the equilibrium.

Our model generates novel implications: When pledgeability increases, the new equilibrium is characterized by fewer banks, and yet, a higher deposit rate – thus, there is a negative correlation between concentration and pricing power (we refer to this as the dichotomy result). We note that several empirical papers provide evidence consistent with this counterintuitive prediction. Even though we allow for price competition and free entry, an improvement in banks' investment opportunities may only be partially passed through to depositors. We derive a novel prediction that the pass-through effect strengthens when pledgeability is low but weakens once pledgeability becomes sufficiently high. In terms of policy implications, there exist an excessively high number of banks in the unregulated banking industry. In this case, the regulator can introduce an upper bound on the number of bank licences to be allocated which would correct the inefficiency.

6 Appendix

6.1 Proofs

Proof of Lemma 1.

Suppose $I = 0$. For any d and R , total bank output is $dp_h X$, of which the pledgeable portion is $\phi dp_h X$, or $\phi p_h X$ per unit. By Assumption A2, $\phi p_h X < 1$, so depositors cannot recover their funds under any contract with $I = 0$ and will never choose such a contract.

Proof of Proposition 2.

We verify each of the five conditions in Definition 1 at the proposed equilibrium (R^*, d^*, λ^*) , and then argue that no other triple satisfies all five.

(E1) Feasibility. By construction, the red dot lies on both the diversion constraint and the

incentive constraint, so both bind. The depositors' participation constraint holds since $R^* \geq R^P$ for $k \geq \bar{k}$ (holding with equality for $k = \bar{k}$). The bankers' participation constraint is slack by Lemma 2, since the incentive constraint binds.

Active-banker optimality. Since the red dot lies on the incentive constraint, active bankers earn $\Pi(d^*, R^*) > 0$ by Lemma 2 and therefore strictly prefer managing a bank to investing in storage.

Depositor optimality. Depositors earn $p_h R^* > 1$, strictly preferring (R^*, d^*) to storage for any $k < \bar{k}$. For $k = \bar{k}$, $p_h R^* = 1$. Under the tie-breaking rule, indifference is resolved in favor of depositing.

(E2) Market clearing. There is a mass m of bankers, of whom a fraction λ^* successfully attract funding, giving $n = \lambda^* m$ active banks, each accepting d^* deposits. Setting $\lambda^* m d^* = 1$ and solving yields:

$$\lambda^* = \frac{(1 - \phi)\Delta X - c}{m[k + c - (1 - \phi)\Delta X]} \quad (18)$$

By Assumption A3, $\lambda^* < 1$, so a positive mass of bankers remain inactive but all depositors are served.

(E3) No profitable entry. Any contract (R', d') with $R' > R^*$ violates at least one of the diversion constraint and the incentive constraint, since the red dot is their intersection and the feasible region lies between them in the (d, R) space. Hence no inactive banker can offer a strictly preferred contract that is also feasible.

Uniqueness. It remains to show that no triple other than (R^*, d^*, λ^*) satisfies (E1)–(E3).

Step 1: Autarky is not an equilibrium. Suppose $\lambda = 0$, so that all depositors invest in the storage technology. In this conjectured equilibrium, an inactive banker can profitably deviate by offering

a contract with $R = R^P + \epsilon$ for arbitrarily small $\epsilon > 0$. By Assumption A1 (in the case $\bar{k} \geq k > \Delta X - c$), there exists a d such that this contract satisfies both the diversion constraint and the incentive constraint, and depositors strictly prefer it to storage. The deviating banker earns strictly positive profits since the participation constraint is slack throughout the feasible region (Lemma 2). This contradicts (E3).

Step 2: The diversion constraint must bind. Suppose the equilibrium (d, R) satisfies the diversion constraint with strict inequality. Then an inactive banker can offer a deposit rate $R + \epsilon$ for arbitrarily small $\epsilon > 0$, attracting depositors away from incumbents under the tie-breaking rule and accepting more deposits. The deviation is feasible for any $R < X - \frac{c}{\Delta}$ and is strictly profitable since the participation constraint remains slack, contradicting (E3). At $R = X - \frac{c}{\Delta}$ the incentive constraint becomes vertical, and no further increase in R is feasible.

Step 3: The incentive constraint must bind. Suppose the equilibrium lies on the diversion constraint but strictly above the incentive constraint – to the left of the red dot in the (d, R) space (for instance, the green dot). By Lemma 2, active bankers earn strictly higher profits than the storage payoff, so an inactive banker has strictly positive profits available by entering. An inactive banker can offer $R + \epsilon$ for arbitrarily small $\epsilon > 0$, attracting depositors away from incumbents, and pushing the equilibrium down the diversion constraint toward the red dot. This contradicts (E3). The equilibrium must therefore lie on both constraints simultaneously, i.e., at the red dot.

Step 4: All depositors are served. Suppose, to the contrary, that $\lambda^* m d^* < 1$, so that some depositors invest in storage. An inactive banker can offer $R = R^P + \epsilon$ to unserved depositors, who strictly prefer this to storage. By the argument in Step 1, this deviation is feasible and strictly profitable, contradicting (E3). Conversely, if $\lambda m d^* > 1$, some active bankers cannot attract d^* deposits and exit at zero cost, restoring market clearing. The unique equilibrium therefore satisfies $\lambda^* m d^* = 1$.

Combining Steps 1–4, the unique equilibrium is (R^*, d^*, λ^*) .

Proof of Lemma 6.

Differentiating d^* with respect to X :

$$\frac{\delta d^*}{\delta X} = -\frac{(1-\phi)\Delta k}{[(1-\phi)\Delta X - c]^2} < 0 \quad (19)$$

Differentiating R^* with respect to X :

$$\frac{\delta R^*}{\delta X} = \frac{\phi k(k+c)}{[k+c-(1-\phi)\Delta X]^2} > 0 \quad (20)$$

Differentiating λ^* with respect to X :

$$\frac{\delta \lambda^*}{\delta X} = -\frac{\delta d^*}{\delta X} \frac{1}{d^{*2}} > 0 \quad (21)$$

Below we provide a graphical proof which illustrates the mechanism. Totally differentiating the diversion constraint with respect to R and X :

$$\left. \frac{dR}{dX} \right|_{DC} = \frac{\phi(1+d)}{d} \quad (22)$$

Totally differentiating the incentive constraint with respect to R and X :

$$\left. \frac{dR}{dX} \right|_{IC} = \frac{1+d}{d} \quad (23)$$

For a given increase in X , both the diversion and incentive constraints shift to the right in the (d, R) space indicating that a higher X leads to an increase in R . To assess the impact of a higher

X on leverage and entry, we compare the magnitude of the lateral shift in the two constraints. The magnitude of the lateral shift in the diversion constraint is smaller than that of the incentive constraint if:

$$\begin{aligned}\frac{\phi(1+d)}{d} &< \frac{1+d}{d} \\ \implies \phi &< 1\end{aligned}$$

The above condition is always satisfied. Given this observation that an increase in X leads to a smaller lateral shift in the diversion constraint than in the incentive constraint, when X increases, the diversion and incentive constraints intersect at a lower point. This implies that a higher X leads to more banks and a lower d .

Proof of Lemma 7.

Differentiating d^* with respect to ϕ :

$$\frac{\delta d^*}{\delta \phi} = \frac{k\Delta X}{[(1-\phi)\Delta X - c]^2} > 0 \quad (24)$$

Differentiating R^* with respect to ϕ :

$$\frac{\delta R^*}{\delta \phi} = \frac{kX \overbrace{(k+c-\Delta X)}^{>0 \text{ by A1}}}{[k+c-(1-\phi)\Delta X]^2} > 0 \quad (25)$$

Differentiating λ^* with respect to ϕ :

$$\frac{\delta \lambda^*}{\delta \phi} = -\frac{\delta d^*}{\delta \phi} \frac{1}{d^{*2}} < 0 \quad (26)$$

Proof of Lemma 8.

Differentiating d^* with respect to k :

$$\frac{\delta d^*}{\delta k} = \frac{1}{\underbrace{(1-\phi)\Delta X - c}_{>0 \text{ by A1}}} > 0 \quad (27)$$

Differentiating R^* with respect to k :

$$\frac{\delta R^*}{\delta k} = \frac{\phi X \overbrace{[c - (1-\phi)\Delta X]}^{<0 \text{ by A1}}}{[k + c - (1-\phi)\Delta X]^2} < 0 \quad (28)$$

Differentiating λ^* with respect to k :

$$\frac{\delta \lambda^*}{\delta k} = -\frac{\delta d^*}{\delta k} \frac{1}{d^{*2}} < 0 \quad (29)$$

Differentiating d^* with respect to c :

$$\frac{\delta d^*}{\delta c} = \frac{k}{[(1-\phi)\Delta X - c]^2} > 0 \quad (30)$$

Differentiating R^* with respect to c :

$$\frac{\delta R^*}{\delta c} = -\frac{\phi k X}{[k + c - (1-\phi)\Delta X]^2} < 0 \quad (31)$$

Differentiating λ^* with respect to c :

$$\frac{\delta \lambda^*}{\delta c} = -\frac{\delta d^*}{\delta c} \frac{1}{d^{*2}} < 0 \quad (32)$$

6.2 Kreps and Scheinkman (1983) with free entry

In this section, we consider the Kreps and Scheinkman (1983) game with capacity constraints and fixed number of firms in the industry. We extend this game to allow for free (costless) entry, and show that, in this extended game, the Bertrand outcome arises as the unique equilibrium.

Consider a simplified version of our model in which there are no informational frictions and the project always succeeds (i.e., $p_h = 1$). Let's now apply the Kreps and Scheinkman (1983) approach to this setting. There are N banks indexed $i = \{1, \dots, N\}$, each demanding deposits, d_i . They face an upward sloping supply curve, $R = a + D$, where a is a positive constant, R is the deposit rate, and D is the total deposits demanded by banks, $D = \sum_{i=1}^N d_i$. Banks use deposits to make loans and the loan rate is exogenously given and constant at X , where $0 < a < X$. This corresponds to the exogenously given marginal cost in Kreps and Scheinkman (1983).

The original game has two stages: In Stage 1 of the game, banks simultaneously and independently choose their capacity to accept deposits, l_i . In Stage 2, each bank observes the other banks' choices of their pre-committed capacities, and they simultaneously choose the deposit rate. We look for the sub-game perfect Nash equilibrium of this game and solve for the equilibrium backwards.

In Stage 2, given capacities, l_i , depositors deposit their funds first to the bank offering the highest deposit rate till this bank's capacity is exhausted. Then, depositors take the residual supply to the bank offering the next highest deposit rate and so on. This iterates till the deposit rate reaches depositors' reservation price. If banks offer the same deposit rate, the market is split equally among them.

In Stage 1, banks set their capacity to accept deposits by anticipating the outcome in the next stage. If the total capacity is higher than the supply of deposits, then banks set the deposit rate equal to the loan rate, X . Thus, to avoid this Bertrand price war, banks choose capacity, l_i to be

equal the Cournot quantities, d^* , thereby softening price competition in the following stage.

A bank j 's profit function is:

$$\Pi_j = (X - R)d_j \quad (33)$$

Using $R = a + D$, and taking the first order condition with respect to d_j :

$$d_j = \frac{X - a - \sum_{i \neq j} d_i}{2} \quad (34)$$

In the symmetric equilibrium, $d_j = d^*$. Substituting, we derive the equilibrium of the Kreps and Scheinkman (1983) game:

$$l_i = d_i = d^* = \frac{X - a}{N + 1}$$

Thus, the total amount of deposits demanded in equilibrium is $D = Nd^*$. Substituting into R :

$$R^* = \frac{NX + a}{N + 1} < X$$

Because banks choose their capacity in the first stage of the game, they are able to earn strictly positive profits even though they compete in deposit rates in the final stage. Anticipating the equilibrium deposit rate that will prevail in the next stage, each bank sets its capacity such that it is exactly filled – there is no excess supply of deposits. As a result, banks can offer a deposit rate strictly below the loan rate, X , since the limited capacity prevents undercutting by competitors from driving the rate up to X .

We now introduce to the Kreps and Scheinkman (1983) game the possibility of bank entry in the first stage of competition. Suppose, as we do throughout the paper, entry is free and costless.

Then, anticipating positive profits, a new bank (indexed $i = N + 1$) enters the market and offers the same capacity in the first stage. This leads to unfulfilled demand in the next stage. Anticipating this, banks adjust their announced capacity. As the number of entrants go up, the equilibrium approaches the Bertrand outcome, and in equilibrium, $R^* = \frac{NX+a}{N+1} = \frac{X+\frac{a}{N}}{1+\frac{1}{N}} = X$ since $N = \infty$.

Proposition 5 (Kreps and Scheinkman (1983) with free entry) *When banks can freely enter the market, the equilibrium reverts to the Bertrand outcome, even if banks can pre-commit to capacity constraints before competing in prices. This implies that pass-through is full, $\frac{\delta R}{\delta X} = 1$.*

The Kreps and Scheinkman (1983) model shows that with a fixed number of firms, pre-committed capacity followed by price competition can yield Cournot-like outcomes. However, once free entry is introduced, firms have an incentive to enter until all economic profits are dissipated. This undermines the disciplining role of capacity constraints, and the outcome reverts to the perfectly competitive Bertrand benchmark.

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